IILM University Greater Noida
Question Bank

Subject Name: DCN
Subject Code: 24CSEC413

(2 Marks Questions)

1. Define data communication and list its components.
2. Differentiate between LAN, MAN, and WAN.
3. What is network topology? Name any four types.
4. State two advantages of layered architecture.
5. List the seven layers of the OSI model.
6. Mention any four differences between OSI and TCP/IP models.
7. What are the responsibilities of the Physical Layer?
8. Define bandwidth and throughput.
9. Compare guided and unguided transmission media.
10. What is switching? Name its types.
11. What are the different types of transmission errors?
12. What is parity checking?
13. Define CRC.
14. What is framing? Why is it needed?
15. Define flow control.
16. What is Stop-and-Wait ARQ?
17. State the working principle of Go-Back-N ARQ.
18. What is Selective Repeat ARQ?
19. Expand HDLC and PPP.
20. What is ARP and RARP?
21. Define different types of computer networks based on geographical coverage.
22. Explain the role of error detection in data communication.
23. Summarize the features of the Point-to-Point Protocol (PPP).
24. Illustrate the layers of TCP/IP protocol suite.
25. Identify the responsibilities of the data link layer in networking.
26. Differentiate between classless and classful addressing.
27. Classify network topologies with suitable examples.
28. Describe the working principle of circuit switching.
29. Recognize different class of IPv4 addressing.
30. List the functions performed by the Network layer in OSI reference model.
31. Illustrate the layers of TCP/IP protocol suite.
32. Identify the responsibilities of the data link layer in networking.
33. Differentiate between classless and classful addressing.
34. Classify network topologies with suitable examples.
35. Describe the working principle of circuit switching with diagram.
36. Recognize different class of IPv4 addressing.
37. Explain Switch, Hub, Router and Gateway.
38. Explain the role of error detection in data communication.
39. Summarize the features of the Point-to-Point Protocol (PPP).
40. Match the following protocols to their OSI layers:
    - HTTP
    - TCP

- IP
- Ethernet

5 Marks Questions

1. Explain different types of networks (LAN, MAN, WAN) with examples.
2. Compare OSI and TCP/IP models in detail.
3. Explain the OSI model with functions of each layer.
4. Describe different network topologies with advantages and disadvantages.
5. Explain responsibilities of the Physical Layer.
6. Compare twisted pair, coaxial cable, and optical fiber.
7. Explain unguided transmission media and their characteristics.
8. Discuss switching techniques (Circuit and, Packet switching).
9. Explain error detection techniques: Parity, Checksum, and CRC.
10. Explain error correction techniques.
11. Discuss framing methods in detail.
12. Explain flow control techniques.
13. Describe Stop-and-Wait ARQ with diagram.
14. Explain Go-Back-N ARQ with suitable example.
15. Describe Selective Repeat ARQ with working mechanism.
16. Explain HDLC frame structure.
17. Explain PPP stack and its components.
18. Discuss multiple access methods: ALOHA and CSMA.
19. Explain IPv4 addressing and subnetting with example.
20. Compare IPv4 and IPv6 addressing schemes.
21. Explain ARP and RARP with working process.
22. Explain Different types of ARP techniques working process.
23. Describe the services provided by the Network Layer.
24. Demonstrate the performance parameters used to evaluate a network.
25. Examine the significance of flow control in data link protocols.
26. Analyze the differences between IPv4 and IPv6 addressing.
27. Implement the Go-Back-N ARQ technique with an example.
28. Evaluate the effectiveness of CSMA/CD and CSMA/CA.
29. A user sends an email. Explain how the data is encapsulated as it moves down the OSI layers with diagram
30. Implement the Go-Back-N ARQ technique with an example.
31. Explain the roles and working principles of the following protocols in a computer network:
    - Internet Protocol (IP)
    - Address Resolution Protocol (ARP)
    - Reverse Address Resolution Protocol (RARP)
32. Explain random access protocols and Controlled Access Protocols.
33. Analyze the differences between IPv4 and IPv6 addressing with its header structure.